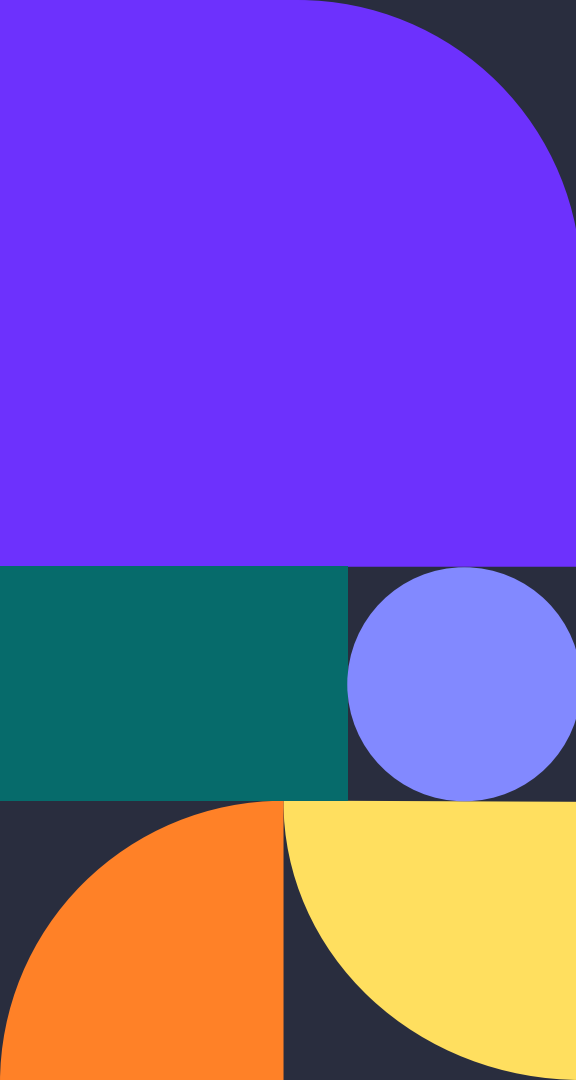
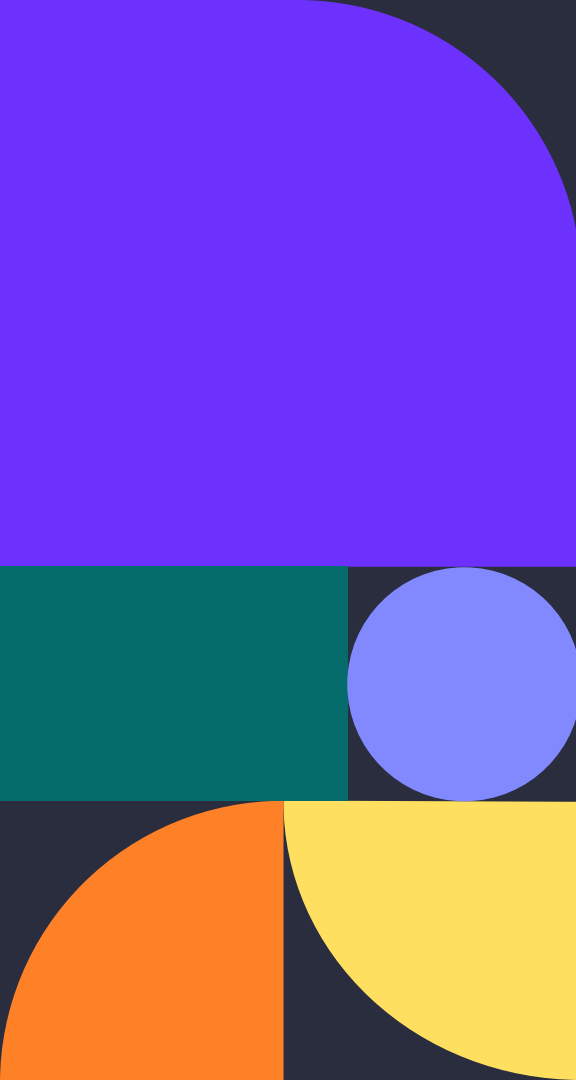




Customer Churn Analysis

By Jonathan J
& Jenna M



Our Data Set

Telco Customer Churn

Contains data about customers including:

- Types of services they signed up for
- Demographic information
- Customer account information
- If the customer left within the last month



What is Customer Churn?

Churn rate, also known as attrition rate, is a business metric that measures the percentage of customers or subscribers who stop using a product or service over a set period.

Goal of the project?

We take the perspective of Telco employees and plan to use techniques from Survival Analysis to identify risky characteristics in our customer base which lead to higher churn. We will also explore how we can use a regression model to better estimate when a customer will churn.



I. Data Preprocessing



Binary Variables:

1. Gender: 0 = female, 1 = male
2. Partner
3. Phone Service - unbalanced binary variable
4. Senior Citizen - unbalanced binary variable
5. Dependents
6. Multiple Lines
7. Online Security
8. Device Protection
9. Streaming TV
10. Paperless Billing
11. Online Backup
12. Tech Support
13. Streaming Movies
14. Churn (Censor Status)

One-hot encoding was applied to these variables.
For Yes/No Variables: 0 = no, 1 = yes.



Categorical Variables:

1. Internet Service: Fiber Optic, DSL, None
2. Contract: M2M, One year, Two year
3. Payment Method: Electronic Check, Mailed Check, Bank Transfer (Automatic), Credit Card (Automatic)

Categorical variables were encoded using C-1 dummy variables.

Continuous Variables:

4. Tenure (Survival Time)
5. Monthly Charges (replaced with log-transform)
6. Total Charges (replaced with log-transform)

Continuous Variables are encoded as either Low (0; below median) or High (1; above median)

Total # of variables: 21

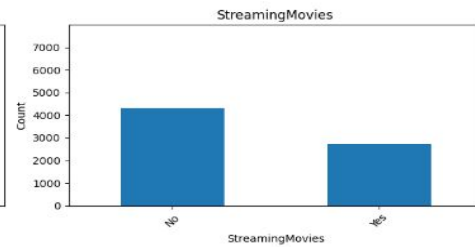
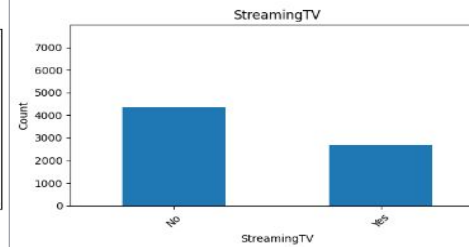
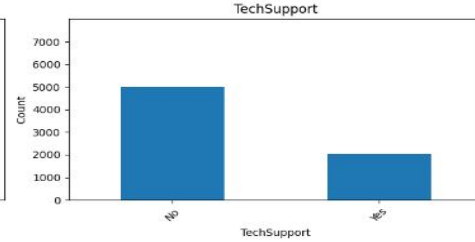
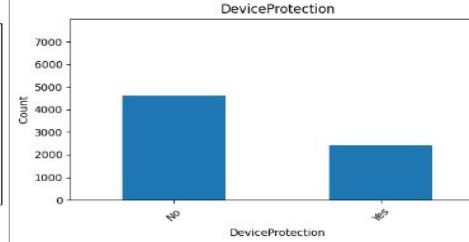
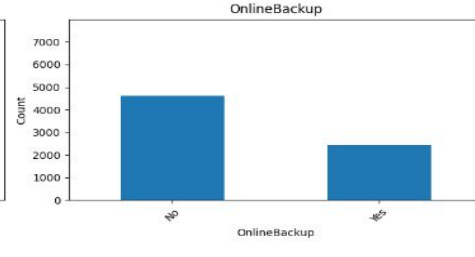
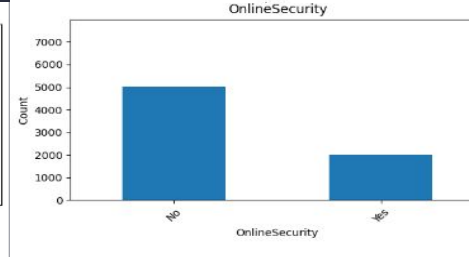
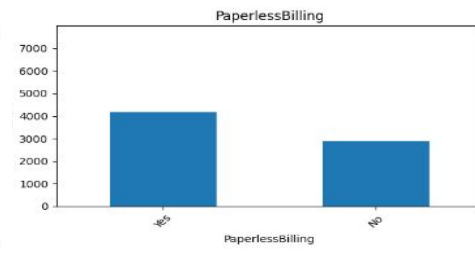
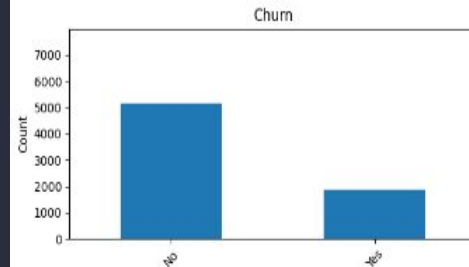
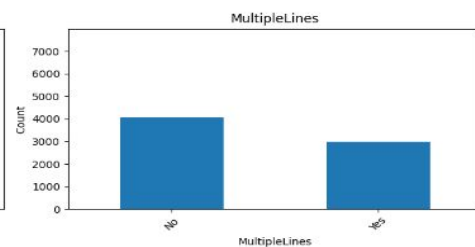
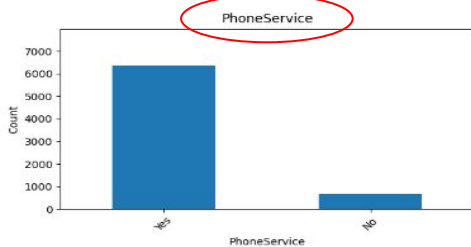
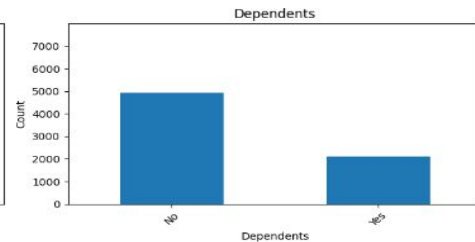
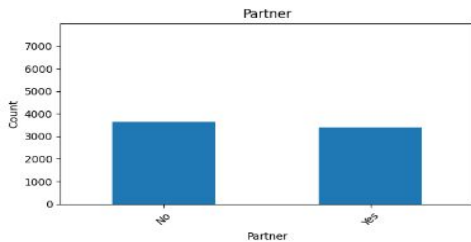
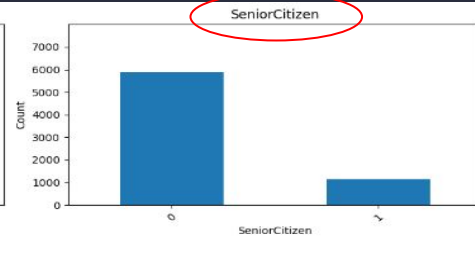
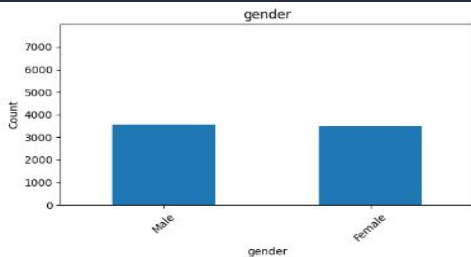


customerID	0
gender	0
SeniorCitizen	0
Partner	0
Dependents	0
tenure	0
PhoneService	0
MultipleLines	0
InternetService	0
OnlineSecurity	0
OnlineBackup	0
DeviceProtection	0
TechSupport	0
StreamingTV	0
StreamingMovies	0
Contract	0
PaperlessBilling	0
PaymentMethod	0
MonthlyCharges	0
TotalCharges	11
Churn	0

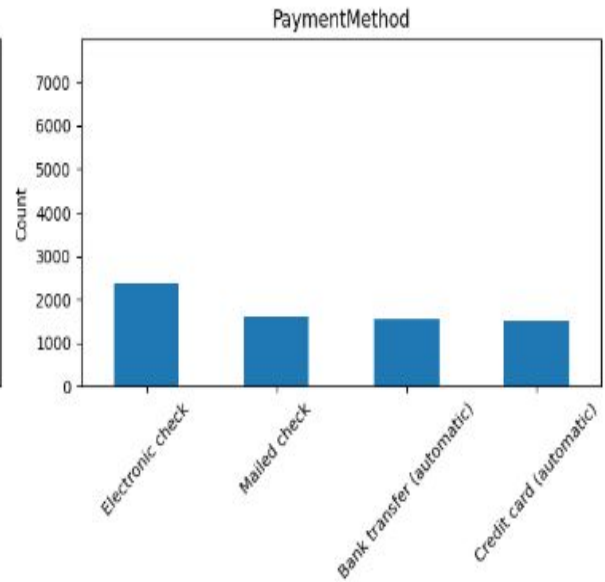
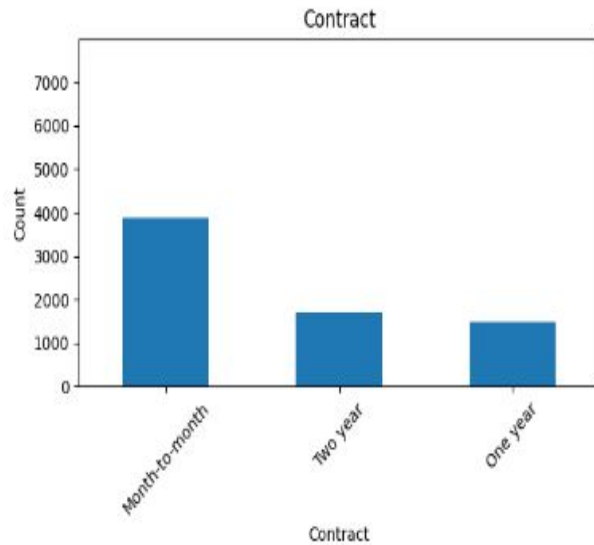
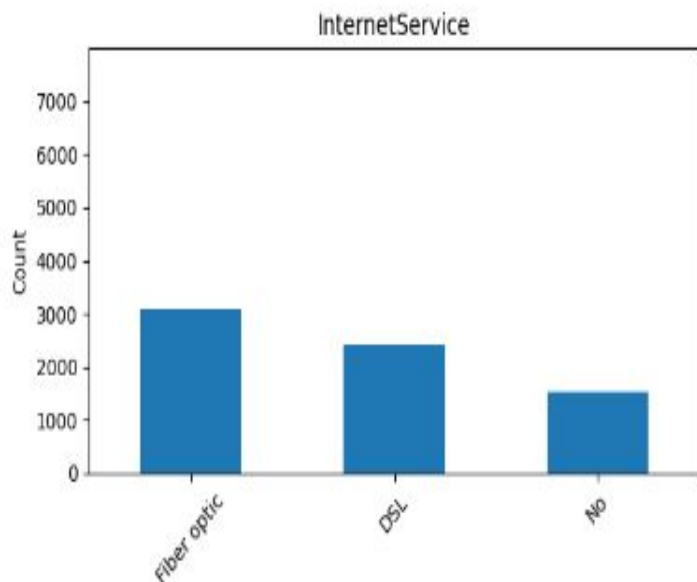
Missing values were replaced with the column median

This is the original list of variables

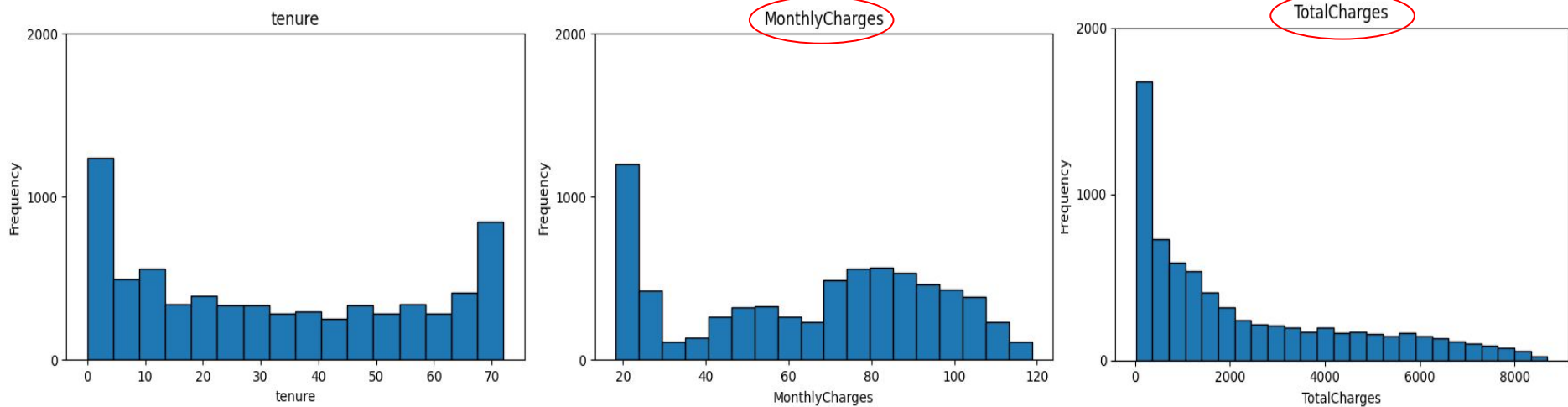
Binary Variables



Categorical Variables

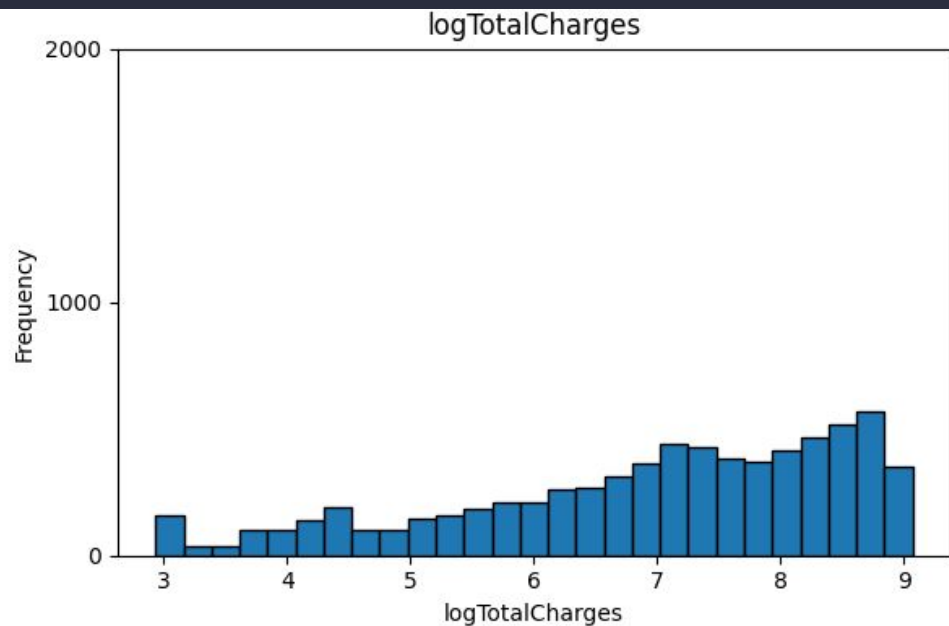
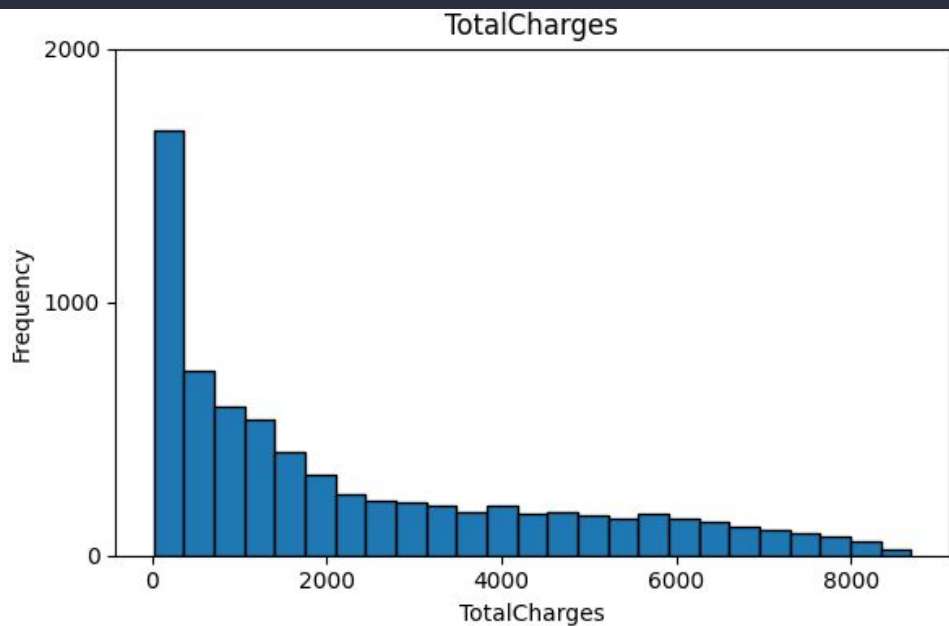


Continuous Variables

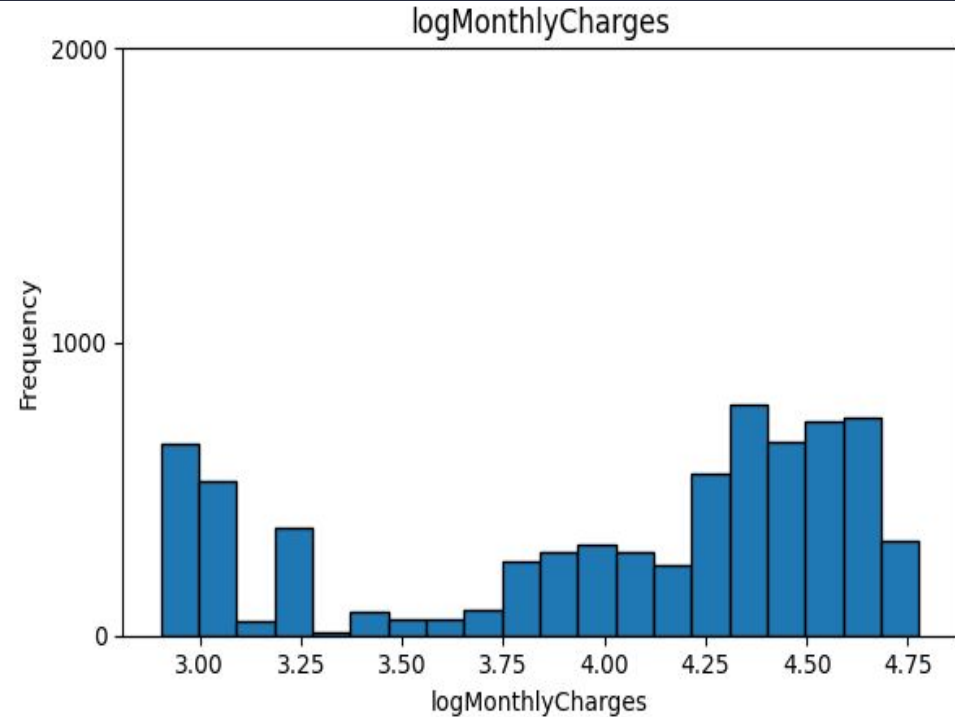
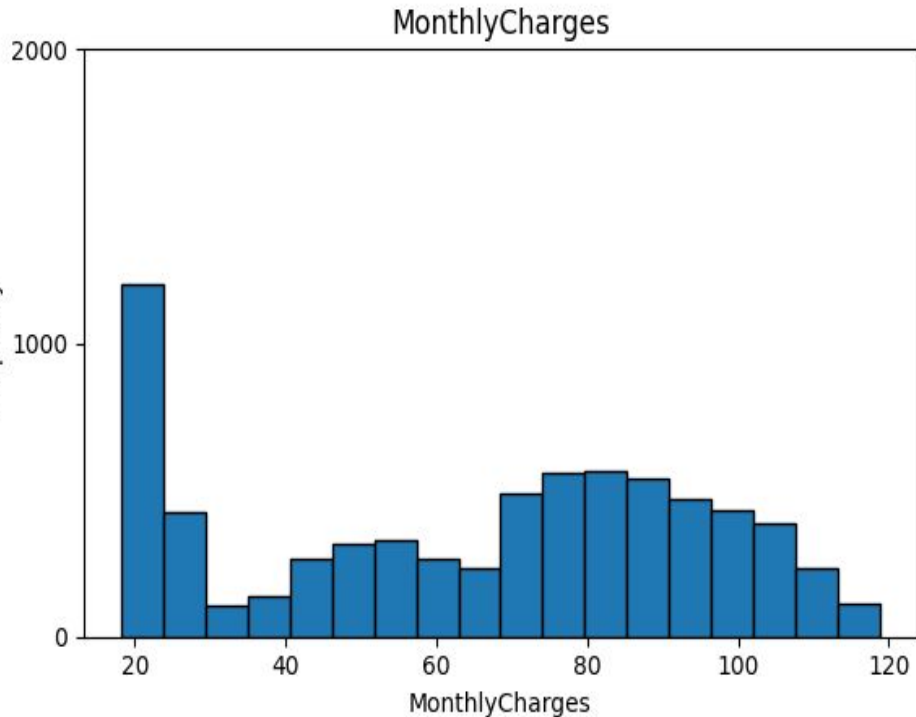


We consider the log-transformed predictors for Charge related data

Log-transformed Total Charges

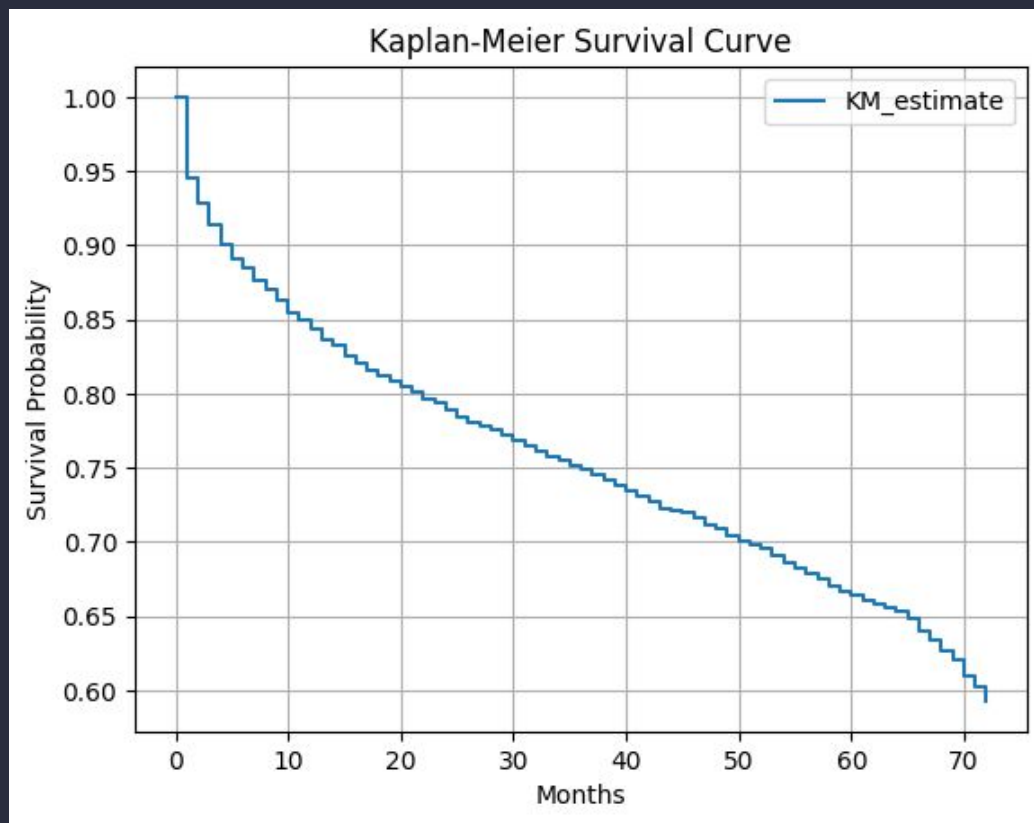


Log-transformed Monthly Charges



A decorative graphic on the left side of the slide, consisting of a yellow quarter-circle on top and a blue quarter-circle on the bottom, both with a radius of approximately 280 units, centered at the origin of the slide's coordinate system.

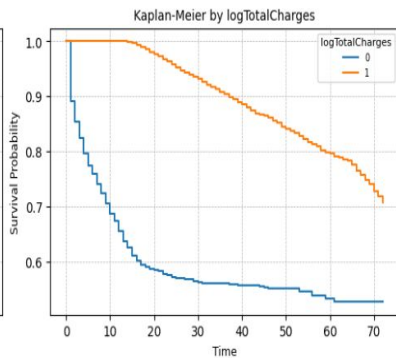
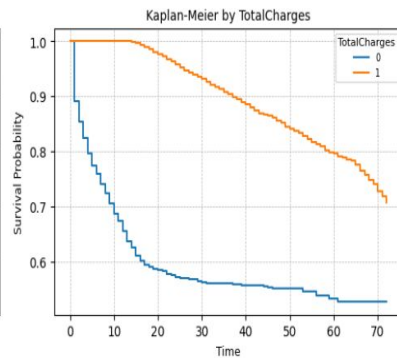
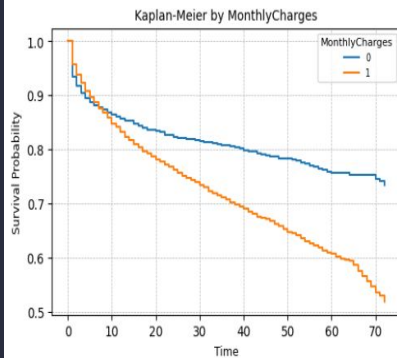
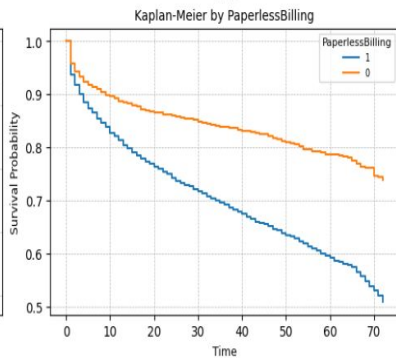
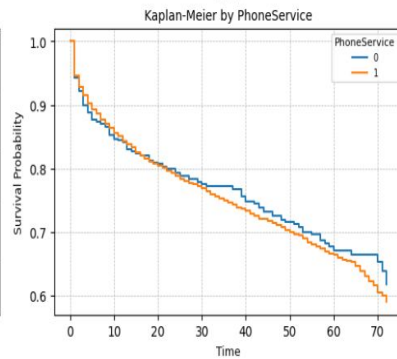
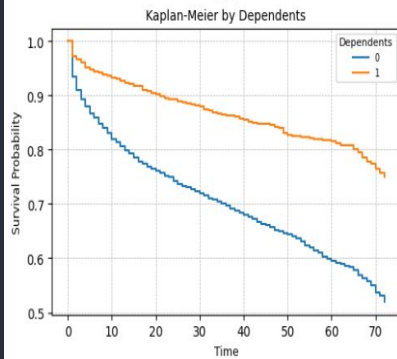
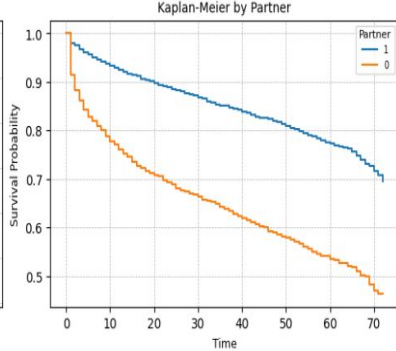
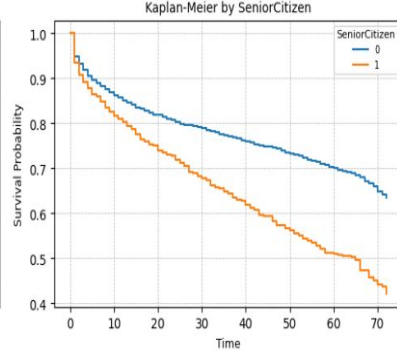
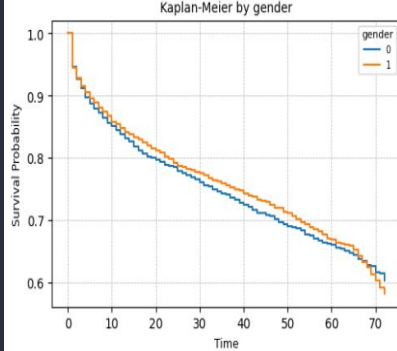
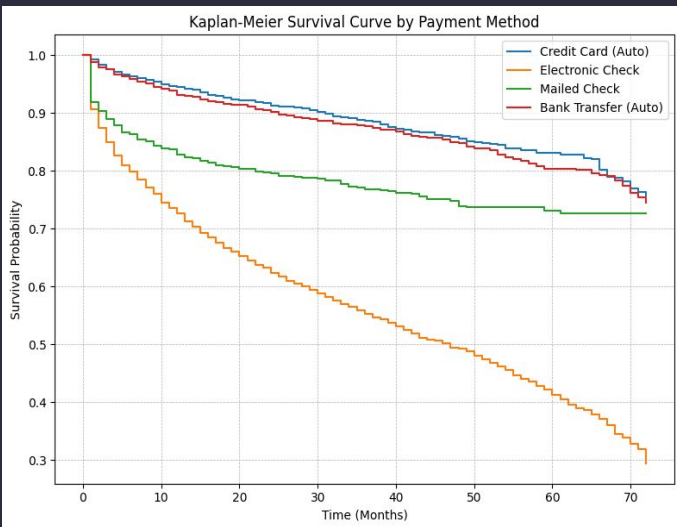
II. Kaplan Meier Plots



Why does this KM plot stop at 0.6?

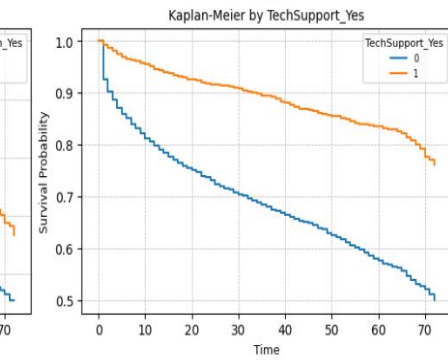
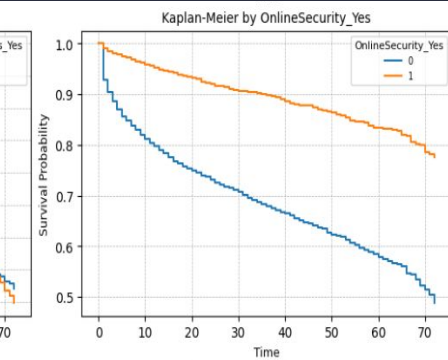
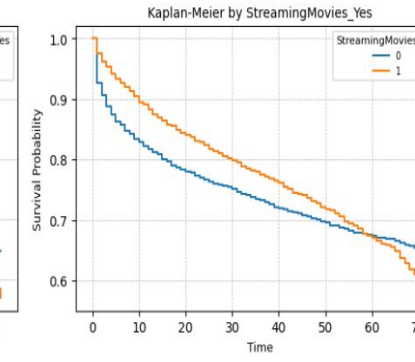
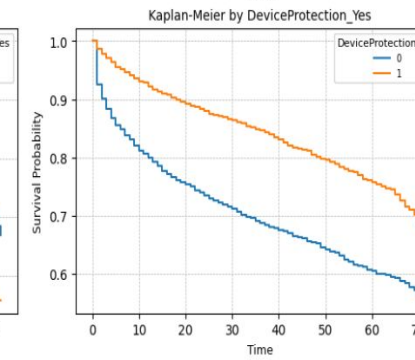
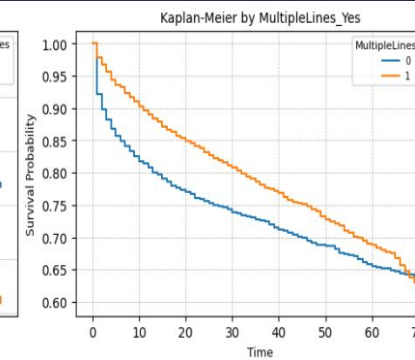
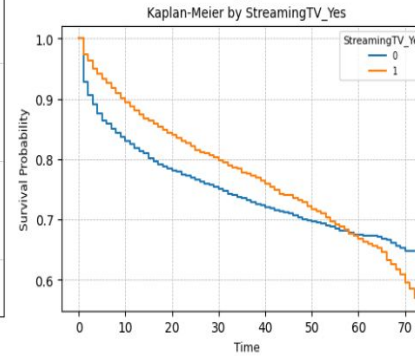
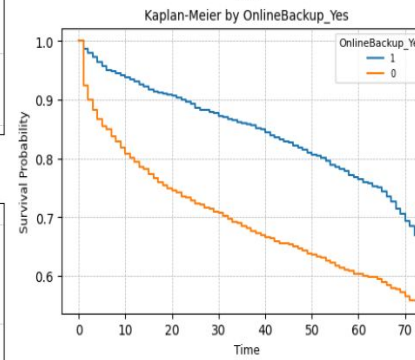
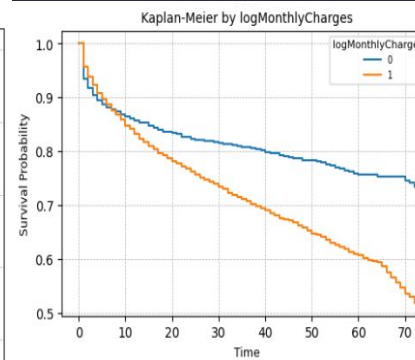
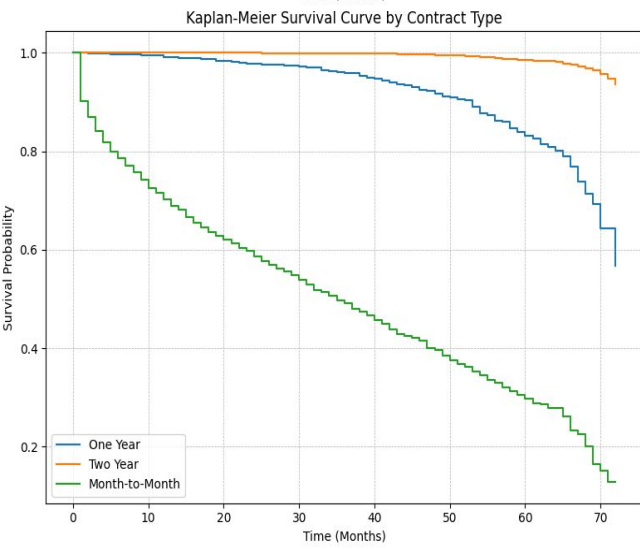
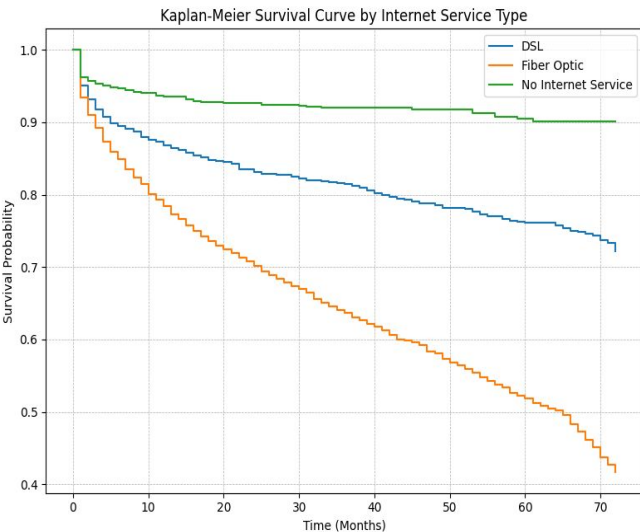
73% of our data is censored (5174/7043)

Therefore, the survival curve will not go to 0 and there is no median survival time.



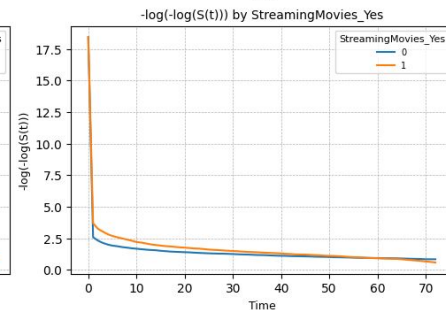
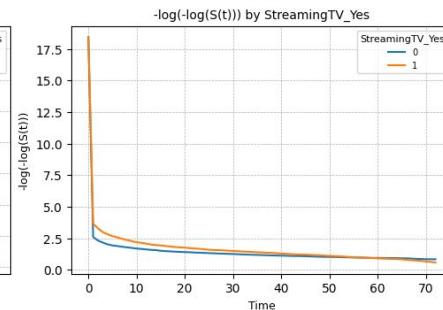
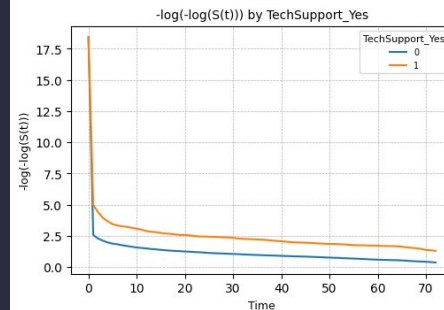
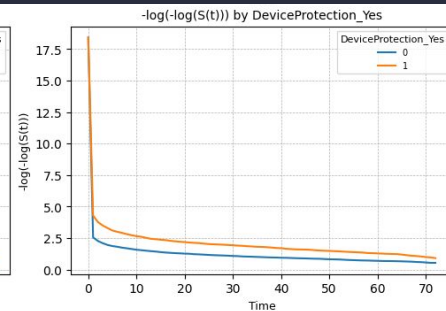
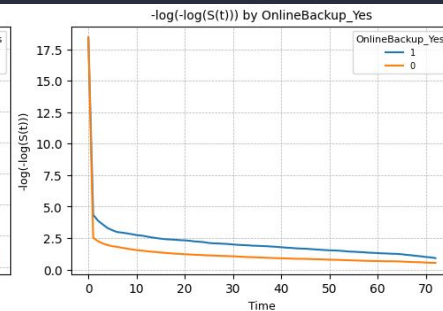
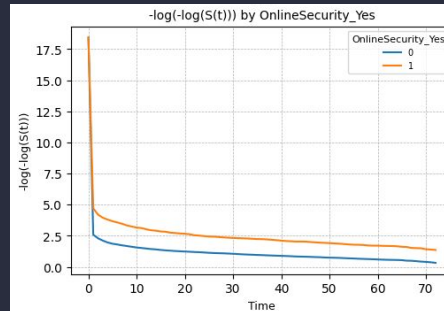
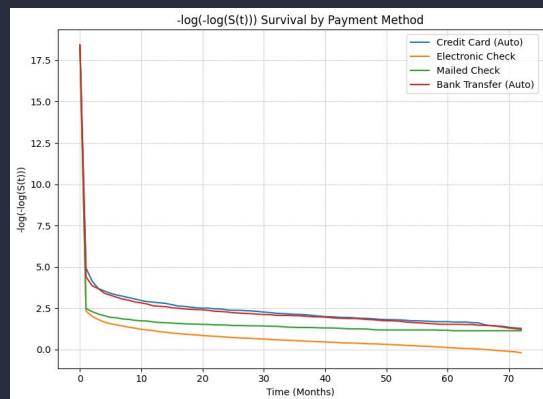
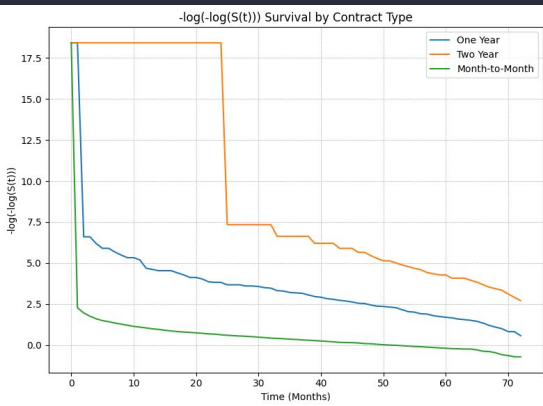
The minimum $S(t)$ for each KM plot differs due to where the censors occur.

Graphs with less censored data will have a larger range of survival probabilities pictured.

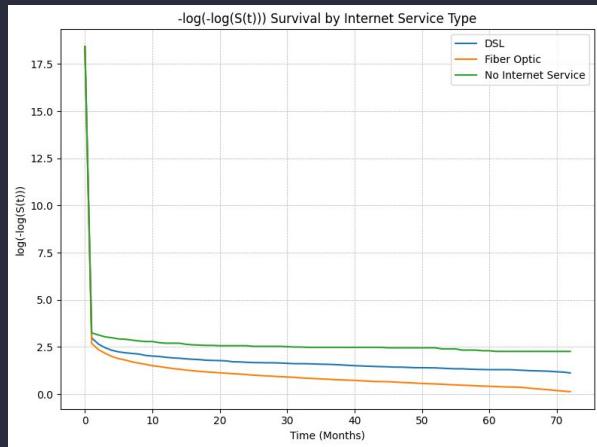




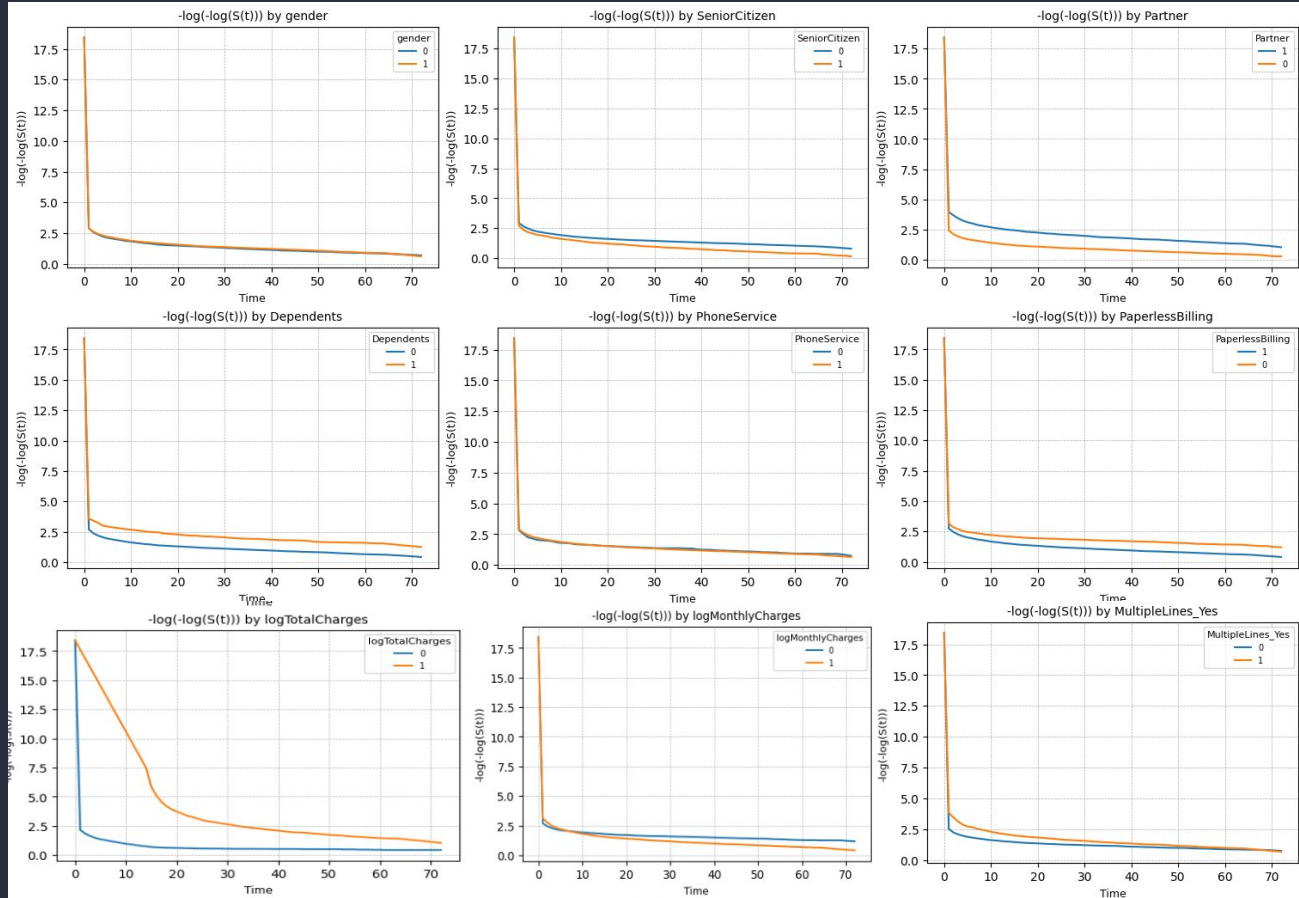
III. Log-Log Plots



Streaming TV and Streaming Movies violated the PH assumptions because the curves cross (are not parallel)



Gender, log Monthly Charges, log Total Charges, Phone Service and Multiple Lines appear to violate the PH assumptions.



Based on a preliminary analysis of the log-log plots, the following variables were chosen for the *no-interaction* regression model:

1. Partner
2. Senior Citizen
3. Dependents
4. Online Security
5. Device Protection
6. Paperless Billing
7. Online Backup
8. Tech Support
9. Internet Service Type
10. Contract
11. Payment Method



IV. Cox PH Regression Model

No-interaction model

model	lifelines.CoxPHFitter											
duration col	'tenure'											
event col	'Churn'											
baseline estimation	breslow											
number of observations	7043											
number of events observed	1869											
partial log-likelihood	-13929.45											
time fit was run	2025-05-13 18:27:23 UTC											
	coef	exp(coef)	se(coef)	coef lower 95%	coef upper 95%	exp(coef) lower 95%	exp(coef) upper 95%	cmp to	z	p	-log2(p)	
Partner	-0.54	0.58	0.05	-0.65	-0.43	0.52	0.65	0.00	-9.86	<0.005	73.80	
SeniorCitizen	-0.12	0.89	0.06	-0.23	-0.01	0.79	0.99	0.00	-2.14	0.03	4.95	
Dependents	-0.04	0.96	0.07	-0.17	0.09	0.84	1.10	0.00	-0.58	0.57	0.82	
OnlineSecurity_Yes	-0.66	0.52	0.07	-0.78	-0.53	0.46	0.59	0.00	-9.91	<0.005	74.47	
DeviceProtection_Yes	-0.36	0.70	0.05	-0.47	-0.25	0.63	0.78	0.00	-6.71	<0.005	35.53	
PaperlessBilling	0.15	1.16	0.06	0.04	0.26	1.04	1.29	0.00	2.59	0.01	6.72	
OnlineBackup_Yes	-0.68	0.51	0.05	-0.78	-0.57	0.46	0.57	0.00	-12.38	<0.005	114.53	
TechSupport_Yes	-0.42	0.66	0.07	-0.55	-0.29	0.58	0.75	0.00	-6.44	<0.005	32.94	
InternetService_Fiber optic	1.26	3.51	0.11	1.03	1.48	2.81	4.39	0.00	11.07	<0.005	92.26	
InternetService_DSL	1.04	2.83	0.12	0.81	1.27	2.26	3.55	0.00	9.02	<0.005	62.25	
Contract_One year	-1.64	0.19	0.09	-1.81	-1.47	0.16	0.23	0.00	-18.68	<0.005	256.35	
Contract_Two year	-3.29	0.04	0.16	-3.61	-2.97	0.03	0.05	0.00	-19.96	<0.005	292.05	
PaymentMethod_Credit card (automatic)	-0.07	0.94	0.09	-0.24	0.11	0.78	1.12	0.00	-0.72	0.47	1.09	
PaymentMethod_Electronic check	0.58	1.78	0.07	0.44	0.72	1.55	2.05	0.00	8.17	<0.005	51.53	
PaymentMethod_Mailed check	0.61	1.85	0.09	0.44	0.79	1.56	2.20	0.00	6.98	<0.005	38.34	
Concordance	0.86											
Partial AIC	27888.89											
log-likelihood ratio test	3447.19 on 15 df											
-log2(p) of ll-ratio test	inf											

Interpretation of Hazard Ratios for the No-Interaction Model

Some examples of HR interpretations:

- The HR for Partner is 0.58, which means that customers with a partner have a 42% lower hazard of churning compared to customers without a partner.
- The HR for Fiber Optic internet service is 2.81 which means the risk of churn is 2.81 times higher for customers with Fiber Optic than customers with no internet service.
- The HR for DSL internet service is 2.83 which means the risk of churn is 2.83 times higher for customers with DSL internet service than customers with no internet service.
- The HR for customers with one year contracts is 0.19 which means that the risk of churn for customers with one year contracts is 81% lower than those who do not have a contract.
- The HR for customers with two year contracts is 0.04 which means that the risk of churn for customers with two year contracts is 96% lower than those who do not have a contract.
- The HR for customers who pay with a mailed check is 1.85 which means the risk of churn is 1.85 times higher than customers who autopay with a bank transfer.

Possible Interaction Terms

Possible interaction terms:

- Type of Internet Service (Fiber Optic, DSL, and no internet service) x Online Security
 - The level of internet quality and add-ons you are paying for may affect how likely a customer is to churn.
- Type of Internet Service (Fiber Optic, DSL, and no internet service) x Tech Support
 - The level of internet quality and add-ons you are paying for may affect how likely a customer is to churn.
- Senior Citizen x Tech Support
 - Senior citizens who pay for tech support may rely on it more, making them less likely to churn compared to those who do not pay for tech support.
- Senior Citizen x Device Protection
 - Senior citizens who also pay for add-ons may be less likely to churn.
- Partner x Dependents
 - Customers who have a family plans may be less likely to churn
- Device Protection x Online Backup
 - Customers who bundle multiple services together may be less likely to churn
- Payment Method x Paperless Billing
 - Those who have to actually view their bill more regularly might be more likely to churn

Interaction Terms which caused Multicollinearity

We received a warning for multicollinearity when we attempted to test for the following interaction terms. Therefore, they were removed:

- Payment Method Electronic Check x Paperless Billing
- Payment Method Credit Card x Paperless Billing
- Internet Service DSL x Device Security
- Internet Service DSL x Tech Support
- Partner x Dependents

Interaction model (Full)

model	lifelines.CoxPHFitter										
duration col	'tenure'										
event col	'Churn'										
baseline estimation	breslow										
number of observations	7043										
number of events observed	1869										
partial log-likelihood	-13926.77										
time fit was run	2025-05-13 19:53:56 UTC										
	coef	exp(coef)	se(coef)	coef lower 95%	coef upper 95%	exp(coef) lower 95%	exp(coef) upper 95%	cmp to	z	p	-log2(p)
Partner	-0.54	0.58	0.05	-0.65	-0.43	0.52	0.65	0.00	-9.86	<0.005	73.70
SeniorCitizen	-0.12	0.89	0.07	-0.26	0.02	0.77	1.02	0.00	-1.74	0.08	3.61
Dependents	-0.04	0.96	0.07	-0.17	0.10	0.84	1.10	0.00	-0.54	0.59	0.76
OnlineSecurity_Yes	-0.70	0.49	0.11	-0.92	-0.49	0.40	0.61	0.00	-6.38	<0.005	32.38
DeviceProtection_Yes	-0.43	0.65	0.07	-0.57	-0.28	0.57	0.75	0.00	-5.87	<0.005	27.75
PaperlessBilling	0.12	1.12	0.06	-0.01	0.24	0.99	1.27	0.00	1.85	0.06	3.95
OnlineBackup_Yes	-0.75	0.47	0.07	-0.88	-0.62	0.41	0.54	0.00	-10.99	<0.005	90.85
TechSupport_Yes	-0.48	0.62	0.11	-0.70	-0.27	0.50	0.77	0.00	-4.36	<0.005	16.23
InternetService_Fiber optic	1.25	3.50	0.11	1.03	1.48	2.79	4.38	0.00	10.91	<0.005	89.63
InternetService_DSL	1.08	2.94	0.12	0.84	1.31	2.32	3.72	0.00	8.97	<0.005	61.53
Contract_One year	-1.64	0.19	0.09	-1.81	-1.47	0.16	0.23	0.00	-18.67	<0.005	256.04
Contract_Two year	-3.30	0.04	0.17	-3.63	-2.98	0.03	0.05	0.00	-19.94	<0.005	291.60
PaymentMethod_Credit card (automatic)	-0.06	0.94	0.09	-0.24	0.12	0.79	1.13	0.00	-0.63	0.53	0.92
PaymentMethod_Electronic check	0.58	1.79	0.07	0.45	0.72	1.56	2.06	0.00	8.24	<0.005	52.37
PaymentMethod_Mailed check	0.54	1.71	0.12	0.30	0.77	1.35	2.16	0.00	4.48	<0.005	17.01
FiberOptic_x_Security	0.08	1.08	0.14	-0.19	0.34	0.83	1.41	0.00	0.56	0.57	0.80
FiberOptic_x_TechSupport	0.09	1.10	0.13	-0.17	0.36	0.84	1.43	0.00	0.68	0.49	1.02
Senior_x_TechSupport	0.03	1.03	0.17	-0.29	0.36	0.75	1.43	0.00	0.20	0.85	0.24
Senior_x_DeviceProt	-0.01	0.99	0.12	-0.24	0.22	0.79	1.25	0.00	-0.08	0.93	0.10
MailedCheck_x_Paperless	0.13	1.14	0.13	-0.12	0.39	0.88	1.48	0.00	1.02	0.31	1.70
DeviceProt_x_Backup	0.21	1.23	0.11	-0.01	0.42	0.99	1.53	0.00	1.87	0.06	4.01
Concordance	0.86										
Partial AIC	27895.54										
log-likelihood ratio test	3452.54 on 21 df										
-log2(p) of ll-ratio test	inf										

Choosing a model (LR Test)

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_6 = 0$$

H_A : At least one of $\beta_1, \beta_2, \dots, \beta_6 \neq 0$

$$LR = -2 \ln L_R - (2 \ln L_F)$$

R = reduced (no-interaction) model

F = full (interaction) model

Likelihood Ratio Statistic: 5.351431001839956

P-value: 0.4995919257146205

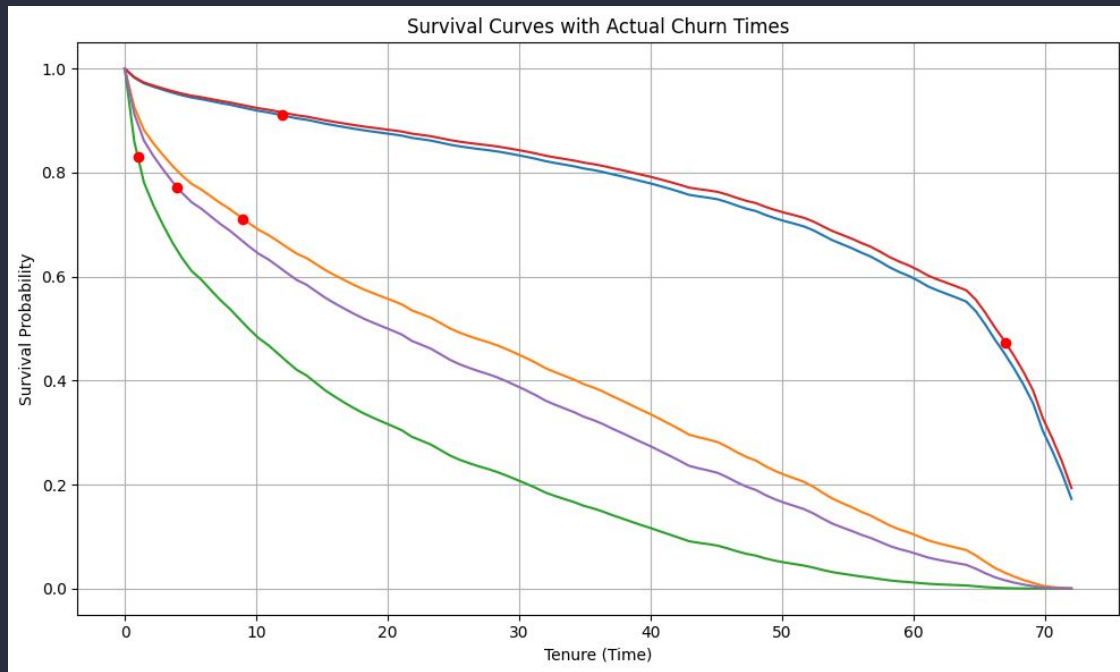
We applied the **LR Test** to determine if our interaction terms were statistically significant.

We've concluded that our interaction terms are **not significant**.



V. Using the Regression Model

Making Predictions using the model



On average, our predictions differ from the median expected Survival times by 16.65 months.

Conclusion

- Our best model included no interaction terms, however we did not consider all possible cases. It is possible that we missed interaction terms which should be included.
- We could have stratified over combinations of groups which did not satisfy the PH assumption.
- Other models such as the Extended model (using time-varying covariates) or Accelerated Failure Times were not considered.

Thank you